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Foundations of Programming

Assignment No: 8

PROBLEM STATEMENT:

Write a program in **C** to accept marks of five courses of a student and compute the result. A student is considered **PASS** if he/she scores **40 marks or more in each course**. If the student passes, calculate the aggregate percentage and assign the grade as follows:

- Aggregate ≥ **75%** : **Distinction**
- Aggregate ≥ **60% and < 75%** : **First Division**
- Aggregate ≥ **50% and < 60%** : **Second Division**
- Aggregate ≥ **40% and < 50%** : **Third Division**

OBJECTIVE:

- To understand the use of **conditional statements** in C
- To apply **logical operators** for decision making
- To calculate **total, percentage, and grade** of a student
- To enhance problem-solving skills using real-life scenarios

THEORY:

In academic evaluation systems, student performance is assessed based on individual subject marks as well as overall aggregate performance. In C programming, such evaluation systems can be implemented using conditional statements like `if`, `else if`, and `else`. Logical operators help in checking multiple conditions simultaneously. Percentage calculation involves arithmetic operations, and grading logic is implemented based on predefined ranges. This program demonstrates structured programming and decision-making in C.

ALGORITHM:

Step 1: Start
Step 2: Declare five variables for marks

m1, m2, m3, m4, m5
and variables for total and percentage.
Step 3: Accept marks of five courses from the user.
Step 4: Check PASS condition
If all marks ≥ 40, then
Go to Step 5
Else
Display "FAIL" and go to Step 9.
Step 5: Calculate total marks
total = m1 + m2 + m3 + m4 + m5
Step 6: Calculate aggregate percentage
percentage = total / 5
Step 7: Assign grade based on percentage

- **If percentage ≥ 75 → Grade = Distinction**
- **Else if percentage ≥ 60 and < 75 → Grade = First Division**
- **Else if percentage ≥ 50 and < 60 → Grade = Second Division**
- **Else → Grade = Third Division**

Step 8: Display PASS, percentage, and grade.
Step 9: Stop.

PLATFORM: Linux

PROGRAM

```
#include <stdio.h>


int main() {

   float m1, m2, m3, m4, m5;

   float total, percentage;


   printf("Enter marks of five courses:\n");

   scanf("%f %f %f %f %f", &m1, &m2, &m3, &m4, &m5);


   /* Check PASS condition */

   if (m1 >= 40 && m2 >= 40 && m3 >= 40 && m4 >= 40 && m5 >= 40) {


      total = m1 + m2 + m3 + m4 + m5;
```

```c
        percentage = total / 5.0;


        printf("\nResult: PASS");
        printf("\nAggregate Percentage = %.2f%%", percentage);


        /* Grade assignment */
        if (percentage >= 75)
            printf("\nGrade: Distinction");
        else if (percentage >= 60)
            printf("\nGrade: First Division");
        else if (percentage >= 50)
            printf("\nGrade: Second Division");
        else
            printf("\nGrade: Third Division");
    }
    else {
        printf("\nResult: FAIL");
    }


    return 0;
}
```

C assignment_09.c > ...

```c
#include <stdio.h>

int main() {
    float m1, m2, m3, m4, m5;
    float total, percentage;

    printf("Enter marks of five courses:\n");
    scanf("%f %f %f %f %f", &m1, &m2, &m3, &m4, &m5);

    /* Check PASS condition */
    if (m1 >= 40 && m2 >= 40 && m3 >= 40 && m4 >= 40 && m5 >= 40) {

        total = m1 + m2 + m3 + m4 + m5;
        percentage = total / 5.0;

        printf("\nResult: PASS");
        printf("\nAggregate Percentage = %.2f%%", percentage);

        /* Grade assignment */
        if (percentage >= 75)
            printf("\nGrade: Distinction");
        else if (percentage >= 60)
            printf("\nGrade: First Division");
        else if (percentage >= 50)
            printf("\nGrade: Second Division");
        else
            printf("\nGrade: Third Division");
    }
    else {
```

```c
            printf("\nGrade: Third Division");
    }
    else {
        printf("\nResult: FAIL");
    }

    return 0;
}
```

INPUT:
The user enters marks of five subjects.
Example Input:
Marks: 78 65 70 80 75

OUTPUT: The program displays the total marks, percentage, result, and grade.

```
Enter marks of five courses:
78 65 70 80 75

Result: PASS
Aggregate Percentage = 73.60%
Grade: First Division
```

CONCLUSION: Thus, the program to calculate student result and grade based on marks was successfully implemented using C programming. This program demonstrates the effective use of conditional statements, logical operators, and arithmetic operations.

FAQs:

1. What is the objective of this C program?
2. Why is an if–else ladder used in this program?
3. Which data types are generally used in this program?